

EXPERIMENT NO- 07

AIM OF THE EXPERIMENT: - TO STUDY THE CONNECTION AND TESTING OF A SINGLE PHASE ENERGY METER WITH UNITY POWER FACTOR LOAD.

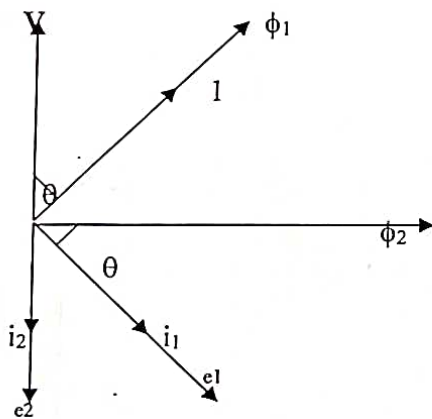
THEORY: -

- Induction type Energy meter (or Watthour meter) is used for single-phase AC measurement only.
- Energy meter is an Integrating type of instruments used to measure the quantity of electric energy supplied to a circuit in a given time.

CONSTRUCTION: -

- It consists of two laminated electromagnets out of which one will be excited by the load current and is called **SERIES MAGNET** or **CURRENT COIL**, while the other is excited by the current proportional to voltage called **SHUNT MAGNET**.
- The shunt magnet is in parallel with the load.
- The alternating flux, produced by series magnets, is directly proportional to the current and is in phase with it while the flux produced by shunt magnet lags behind the voltage by 90° .
- The series magnets consist of few turns of thick copper wires and shunt magnets has more number of turn of thin copper wire.
- A thin Aluminum disc is mounted between the two magnets. So it cuts the leakage fluxes of both magnets.
- The function of the **LAG ADJUSTER**, is shunt coil flux lags behind the voltage exactly by 90° . this is accomplished by making the shunt coil inductive as possible by adjustment of copper ring fitted on the central limb of shunt magnets.
- The instrument is spring controlled. The springs are fixed to the spindle of the rotating aluminum disc having a pointer.

OPERATION: -



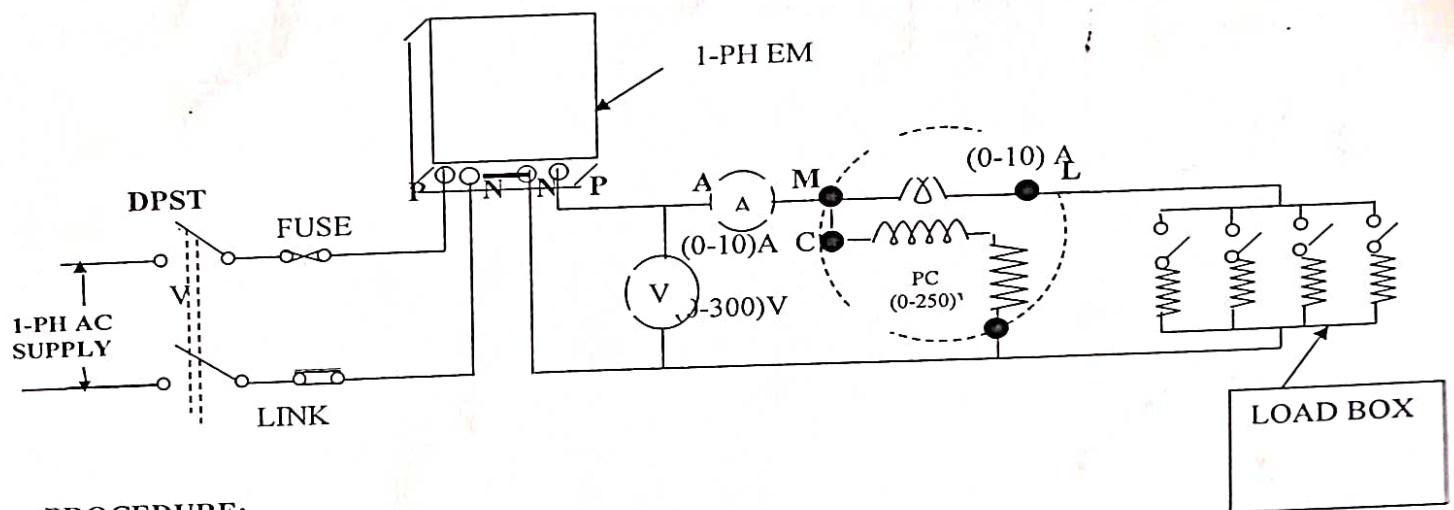
- The shunt magnets produces a flux ϕ_2 lags behind the applied voltage by exactly 90° .
- The series magnet produces a flux ϕ_1 . Which is in phase current (I).
- The flux ϕ_2 and ϕ_1 induces an emf. e_2 and e_1 respectively in the disc, lags behind the respective fluxes by 90° .
- The eddy currents i_2 and i_1 set up by these induced emf in phase with the respective.
- The Aluminum disc is thus acted upon by two torques which are oppositely directed one due to interaction of i_1 and ϕ_2 & other due to interaction of i_2 and ϕ_1 .
- From the vector diagram, (i) the angle between ϕ_2 and i_1 is θ (ii) angle between ϕ_1 and i_2 is $(180-\theta)$.

- $T_1 = k_1 * \phi_2 * i_1 * \cos 0$ & $T_2 = k_2 * \phi_1 * i_2 * \cos (180 - 0)$
- Average torque acting on the disc $T_d = k_1 * \phi_2 * i_1 * \cos 0 - k_2 * \phi_1 * i_2 * \cos (180 - 0)$
- $= k_1 * \phi_2 * i_1 * \cos 0 + k_2 * \phi_1 * i_2 * \cos 0$
- But $i_2 \propto V$ and $\phi_2 \propto V$, $i_1 \propto I$ and $i_2 \propto I$
- $T_d = VI \cos 0 \propto \text{Power}$ and if N is the steady speed
- The Braking Torque $T_b = k_4 N$
- The average torque acting on the Aluminum disc is proportional to the **POWER** consumed by the circuit.
- Total number of revolution $= \int N \cdot dt = \int k \cdot VI \cos 0 \cdot dt = k \int (\text{power}) \cdot dt$. It represents the energy consumed

APPARATUS REQUIRED: -

SL. NO.	NAME OF THE APPARATUS	SPECIFICATION	QUANTITY
1	VARIAC	1-PH, (0-270) V, 5A	01
2	VOLTMETER	(0-300) V, MI TYPE	01
3	AMMETER	(0-5) A, MI TYPE	01
4	ENERGYMETER	(5-10) A, 1-PHASE, 900 RPU	01
5	LOAD BOX	RESISTIVE LOAD	01
6	CONNECTING WIRES	3/22 SWG, INSULATED CU WIRE	AS REQUIRED

CIRCUIT DIAGRAM: -



PROCEDURE: -

1. Make the connection as per the circuit diagram.
2. Note down the initial reading of the Energymeter.
3. Switch on the supply.
4. Switch on the load for 5mins.
5. Take the reading of the wattmeter and Energymeter.
6. Increase the load.
7. Repeat steps 2-5 for 10 mins and 15 mins.
8. Switch off the supply and find out the error.

CONCLUSION: - FROM THE ABOVE EXPERIMENT, WE HAVE STUDIED THE CONNECTION AND TESTING THE ENERGY METER AT UNITY POWER FACTOR WITH.....% OF ERROR.

PRECAUTIONS: -

1. All the connection should be right & tight.
2. The voltmeter & ammeter should be carefully chosen so that their range are more than the maximum value to be measure.
3. The field current should not be increased to cross the rated value.
4. Supply should be switched-on after ensuring correctness of connections.
5. Any live terminal should not be touched while supply is on.

QUESTIONS: -

1. What is an Enegymeter? What is the unit of Electrical energy?
2. What do you mean by one unit of energy?
3. What do you mean by Energy meter constant? How do you determine its value?
4. What type of load will be used to calibrate the energy meter at U.P.F.?
5. Will it be possible to calibrate the meter, if 0.5 Lag or Lead using resistive load and three-phase supply?
If it is possible, explain how it can be achieved?